Wolf Pack Habitats Threats and Suitability in Washington State

Introduction

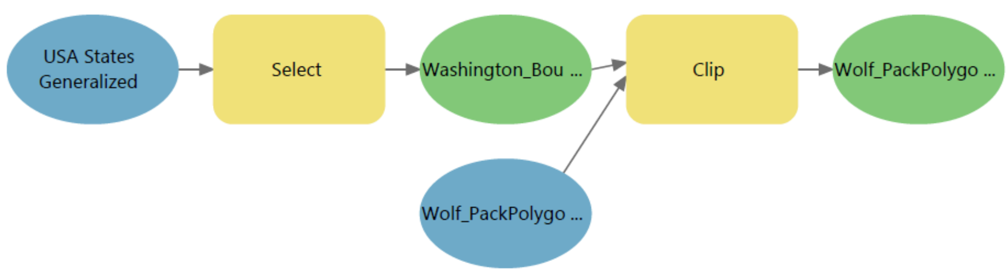
For this GIS project, I focused on the habitat and conservation of wolf packs within Washington State. The objective is to use spatial analysis to investigate three primary questions:

1. Where are the current habitats of wolf packs in Washington State?
2. What are the potential threats to these habitats?
3. How suitable are different areas in Washington for wolf habitation?

By analyzing these questions, we can better understand wolf pack distribution and develop strategies for coexistence between wolves and human activities. This topic is of particular interest because wolves play a crucial role in maintaining ecological balance, but their survival is often threatened by human activities and environmental changes. The spatial nature of this study involves analyzing factors such as land ownership, proximity to roads, and elevation to determine where wolves are most likely to thrive and where they face the greatest risks.

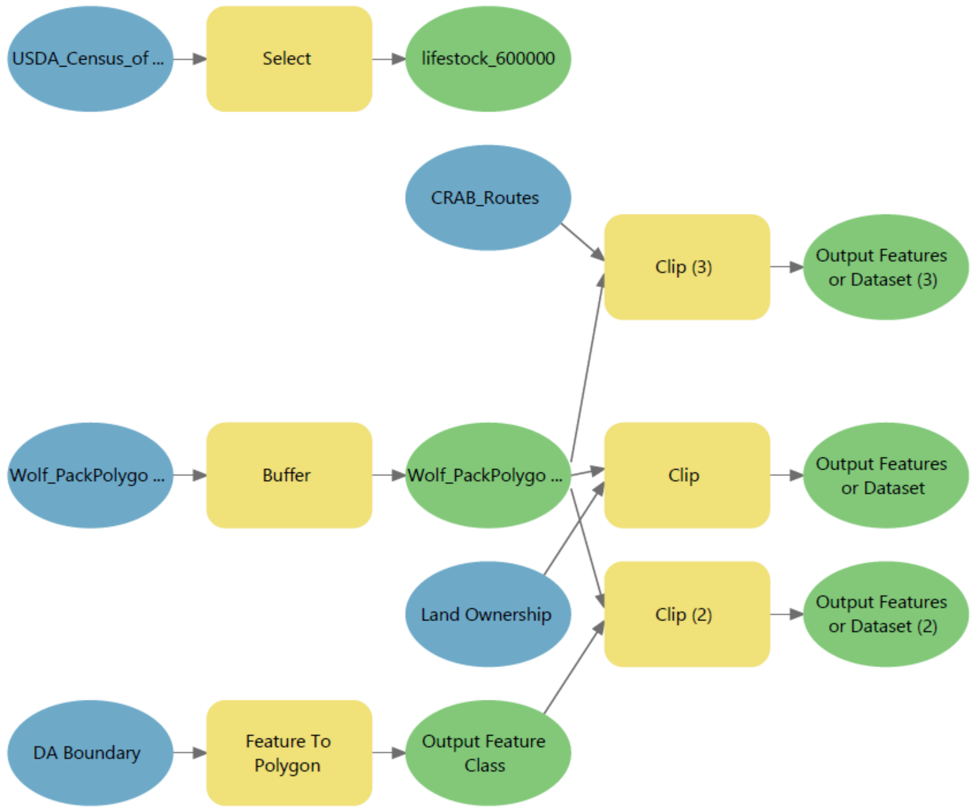
Methods

Map 1: Habitat of Wolf Packs This map illustrates the current habitats of wolf packs across Washington State. Utilizing spatial data on wolf pack locations, I mapped their distribution and used vector data to define the primary habitat areas of these packs. The data was clipped to focus exclusively on Washington State, ensuring a region-specific analysis. Additionally, I incorporated a layer showing various habitat types within the state. The analysis reveals that wolf packs predominantly reside in forested areas, highlighting their preference for these environments.



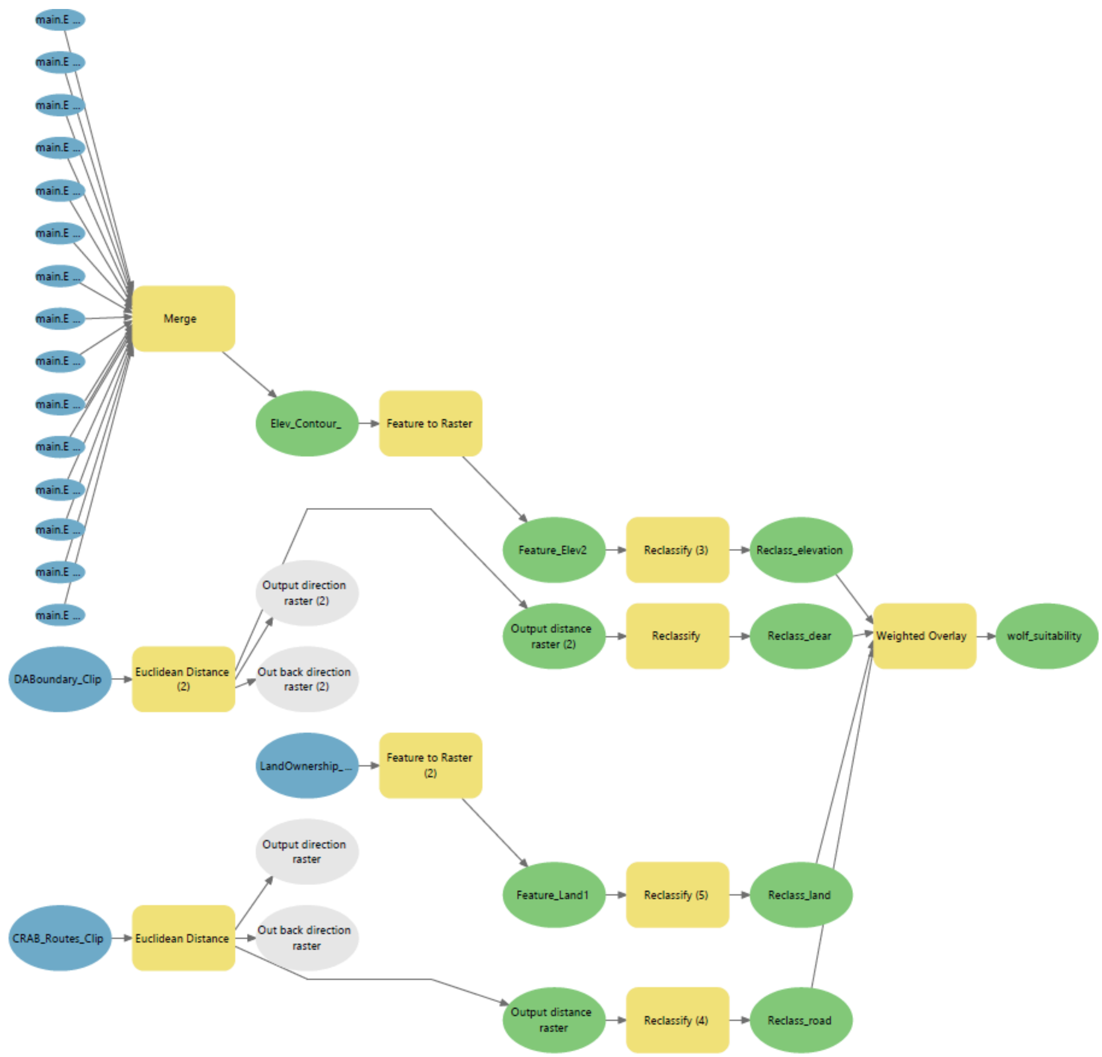
Map 2: Potential Threats to Wolf Packs This map identifies potential threats to wolf pack habitats in Washington State by incorporating data on human activities such as roads, deer hunting areas, and livestock farms. These factors are known to negatively impact wolf populations. A layer of land ownership within wolf pack areas was also included to assess potential human-related threats. Using vector analysis, I performed spatial overlays to determine the proximity of wolf habitats to these threats. To simulate the potential expansion of wolf activity, I applied buffer zones of 5 miles around current wolf pack locations. Areas with a high intensity of livestock farming (over 600,000 livestock) were selected to evaluate their impact on wolf packs. Additionally, I analyzed land ownership, focusing on American Indian lands where illegal hunting of wolves might occur due to traditional practices. Other significant threats identified include legal deer hunting areas close to wolf pack activities and roads within wolf

habitats, which could lead to road accidents and habitat fragmentation. The deer hunting areas, roads, and land ownership data were clipped within the buffered wolf pack zones, and the layers were overlaid to create the final map.



Map 3: Wolf Pack Habitat Suitability The third map evaluates the habitat suitability for wolf packs across Washington State by integrating various environmental and human-related factors. This analysis considers elements such as elevation, proximity to roads, land ownership, and deer hunting areas, which were analyzed in Map 2.

Due to the fragmented nature of the available elevation data, I merged the vector data into a comprehensive dataset that covers the wolf pack areas. I then converted these vector layers into raster data, including land ownership. For deer hunting areas and roads, I utilized Euclidean distance analysis, transforming these factors into raster layers. Each factor was reclassified on a scale of 1 to 10, with 1 representing low suitability and 10 indicating high suitability for wolf activity. Elevations below 8,000 feet received the highest score, while American Indian lands were assigned lower scores due to the potential threat of illegal hunting. For deer hunting areas and roads, scores increased with greater distance from these features, indicating safer habitats. Finally, I use a weighted overlay of the reclassified raster layers to produce an overall suitability score for wolf habitats. This map provides a comprehensive view of the areas most suitable for wolf packs, aiding in the identification of regions where conservation efforts can be prioritized.



Coordinate System Used

- WGS 1984 Web Mercator - Used for its ability to accurately display the broader landscape across varying elevations.

Data Sources

- **Wolf Pack Locations, Deer Hunting Area, Land Cover data, Land Ownership:** ArcGIS Online
- **Roads:** Washington Geospatial Data Portal
- **Live Stock Farms:** WSDA
- **Elevation:** USGS

Results/Discussion

The analysis of wolf pack habitats, potential threats, and habitat suitability in Washington State provided critical insights for conservation and management efforts. The first map successfully identified the current habitats of wolf packs, revealing that these animals predominantly reside in forested areas. This map established a foundational understanding of where wolf populations are concentrated.

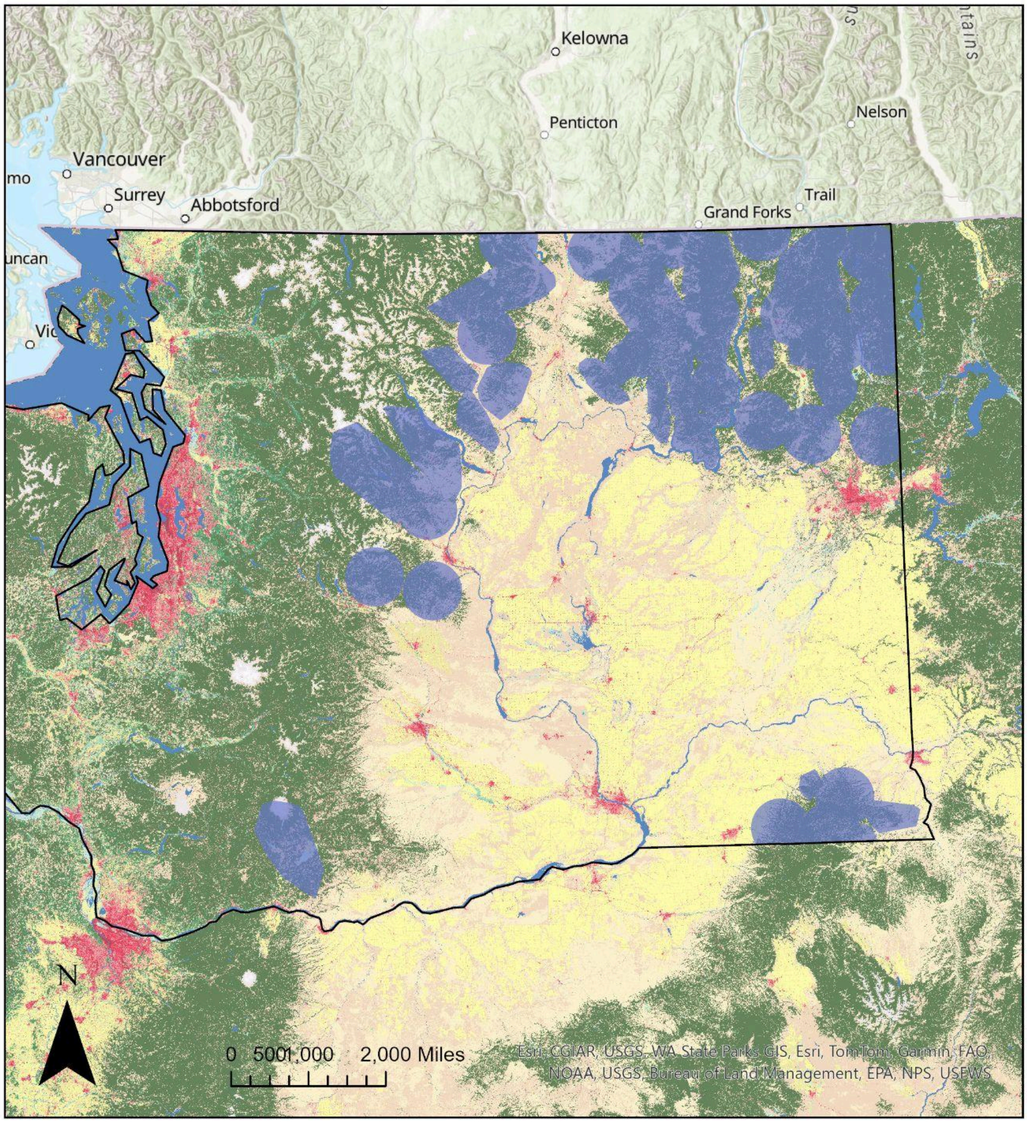
The second map highlighted potential threats to wolf packs by overlaying human activities such as livestock farming, roads, and deer hunting areas. It also revealed that large American Indian Land areas could pose a threat due to traditional hunting practices. Buffer zones were used to simulate the possible expansion of wolf activity, and the analysis pinpointed regions where wolves are most at risk from human encroachment.

The third map, which assessed habitat suitability, integrated environmental factors like elevation with the human-related threats identified in the second map. I encountered significant challenges with this map due to the large size of the elevation data, which caused ArcGIS to crash multiple times during the merging process. This experience underscored the need to find more efficient data sources or consider a smaller study area for future projects. Despite these difficulties, the final suitability map provided a detailed overview of areas most conducive to wolf habitation, balancing both natural and anthropogenic factors.

Throughout this project, I made several cartographic decisions to ensure that the maps were both informative and accessible. The choice of coordinate systems was influenced by the need for precision in regional analysis and the desire to provide clear visualizations across varying scales. Color schemes and symbol choices were carefully selected to differentiate between various data layers, making the maps easier to interpret. Additionally, I adjusted the transparency of each layer to allow for effective overlay and interpretation of multiple data sources simultaneously.

Future studies could benefit from incorporating more detailed data on wolf prey species and refining the spatial analysis to include seasonal variations in habitat use. Overall, this project has enhanced my understanding of the complexities involved in wildlife management and the critical role of spatial analysis in conservation efforts.

Study Area and Wolf Habitat Distribution



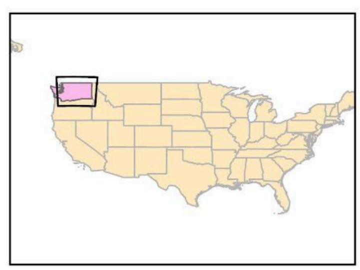
Washington_Boundary
Wolf_Packs

USA NLCD Land Cover

ClassName

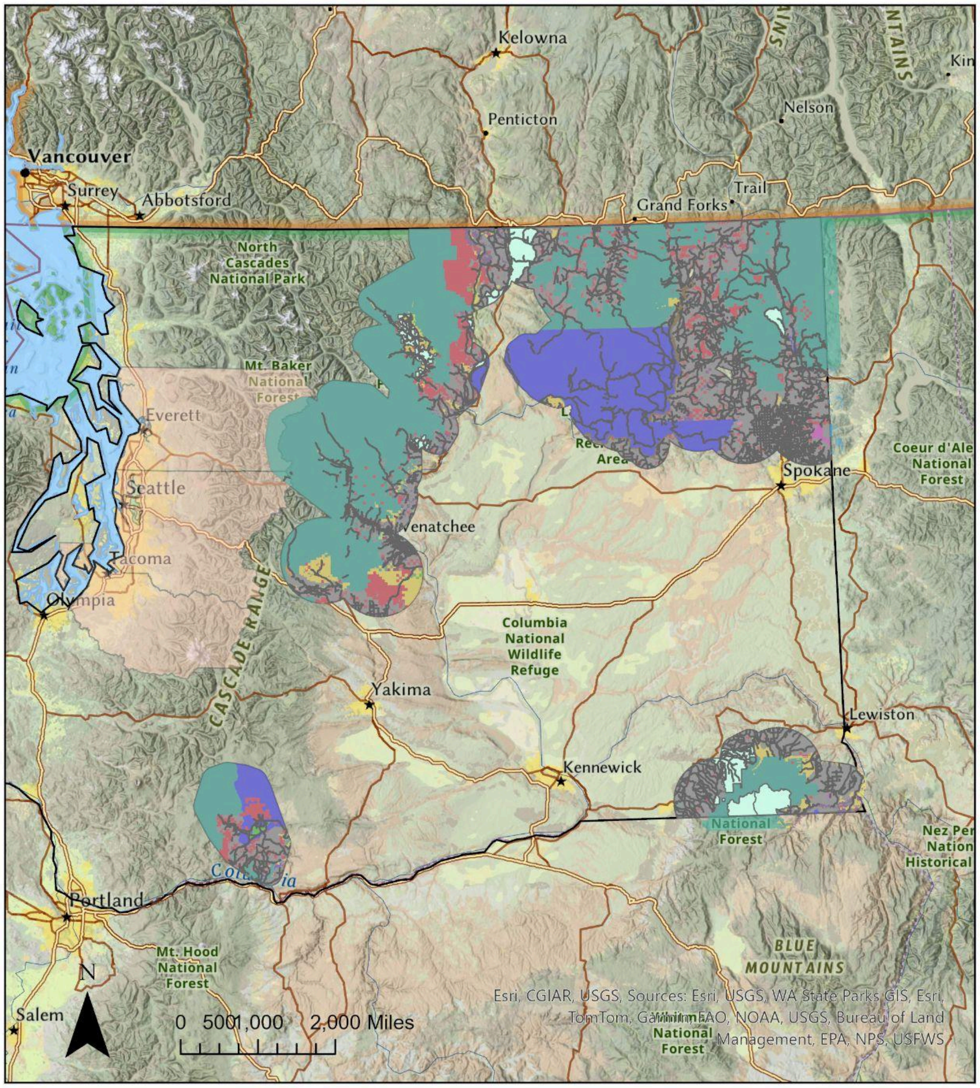
- Open Water
- Perennial Snow/Ice
- Developed Open Space
- Developed Low Intensity
- Developed Medium Intensity
- Developed High Intensity
- Barren Land
- Deciduous Forest
- Evergreen Forest
- Mixed Forest
- Dwarf Scrub
- Shrub/Scrub
- Grassland/Herbaceous
- Sedge/Herbaceous
- Lichens
- Moss
- Pasture/Hay
- Cultivated Crops
- Woody Wetlands
- Emergent Herbaceous Wetlands

Overview of key Habitats and environmental features within the wolf packs region



Susan Wu, ESRM250,
Final Project Map #1
data provided by Esri, USGS

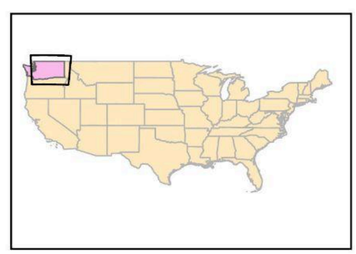
Potential Threats to Wolf Packs



roads
dear_hunting_area
livestock_population_600000
wolf_pack_5mi
Washington_Boundary

LandOwnership
Forest Service
American Indian Lands
State Department of Natural Resources
State Fish and Wildlife
Bureau of Land Management
U.S. Fish and Wildlife Service
National Park Service
State Park and Recreation
State Department of Land
Non-Governmental Organization
City Land

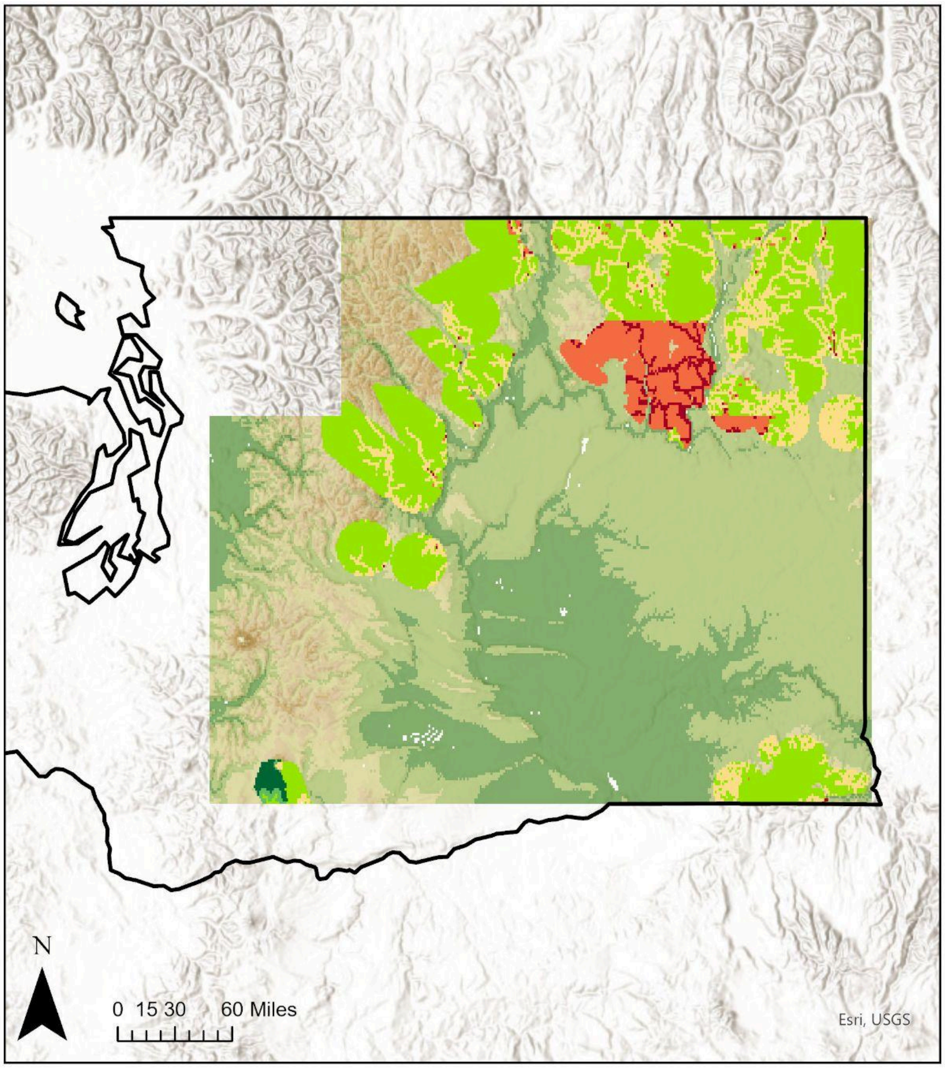
This map illustrates the potential threats to wolf populations within the study area, including the proximity of livestock farms, designated deer hunting zones, major roads, and areas of land ownership.



I am mainly concerned about the regions of American Indian land where traditional hunting practices may pose a risk to wolf populations due to potential illegal hunting activities.

Susan Wu, ESRM 250,
Final Project Map #2
data provided by Esri, USGS

Wolf Habitat Suitability Analysis



This map integrated land Ownership, road proximity, deer hunting area proximity, and elevation to identify optimal areas for wolf pack settlements

Washington_Boundary

wolf_suitability

Value

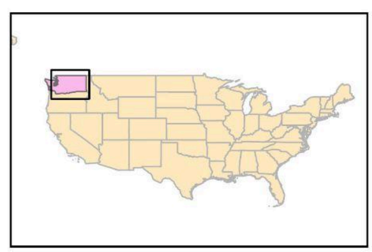
- 5
- 6
- 7
- 8
- 9
- 10

Elevation

VALUE

- 20.001 - 1,555.556
- 1,555.557 - 3,091.111
- 3,091.112 - 4,626.667
- 4,626.668 - 6,162.222
- 6,162.223 - 7,697.778
- 7,697.779 - 9,233.333
- 9,233.334 - 10,768.889
- 10,768.89 - 12,304.444
- 12,304.445 - 13,840

Washington_Boundary



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Final Project Map #3
data provided by Esri, USGS